

How biosim-lab Actually Works

The biology, the physics, the maths and the code —
explained for people who are not any of those things

A companion to the biosim-lab source code

September 7, 2026

Who this is for. Someone who has heard the words “cancer cell”, “sound wave” and “simulation” and would like to know, honestly and in order, what this software does and why anyone should believe it. No physics, no biology and no programming are assumed. Every equation is preceded by a sentence saying what it means in words, and every symbol is defined in a table right underneath. Nothing is left as “it can be shown that”.

What it is not. It is not a summary of results. Roughly a fifth of this document is about the things the model gets wrong, because that is the part a reader actually needs in order to decide whether to trust the other four fifths.

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Part I — The problem, which is a biology problem

1 Finding one cell in five billion

A person with an advanced solid tumour sheds cells from that tumour into the bloodstream. These are called circulating tumour cells, or CTCs. They matter for three reasons: their number tracks how the disease is going, they can be sequenced to see which mutations the tumour carries *today* rather than on the day of the biopsy, and getting them requires only a blood draw instead of a needle in an organ.

The difficulty is arithmetic. In one millilitre of blood there are roughly:

Cell	Number per mL	What it does
Red blood cells (erythrocytes)	5 000 000 000	carry oxygen
Platelets (thrombocytes)	250 000 000	clot wounds
White blood cells (leukocytes)	7 000 000	immune system
Circulating tumour cells	about 1	the thing you want

In plain English

You are looking for one particular cell among about five billion others. If the blood cells were grains of sand, you would be searching a beach roughly the size of a tennis court for one specific grain — and you have to find it without damaging it, because you want to grow it and sequence it afterwards.

There are two families of approach.

Chemical. Coat some magnetic beads with an antibody that sticks to a protein found on tumour cells but not on blood cells, mix, and pull the beads out with a magnet. This works and is the basis of the approved clinical tests. Its weakness is that it only catches cells carrying the protein you chose, and the most dangerous tumour cells — the ones that have changed their identity in order to travel — are exactly the ones that switch that protein off.

Physical. Ignore chemistry entirely. Tumour cells are simply *bigger* than blood cells. Sort on size. Nothing can be switched off to evade a size filter.

This software simulates a physical sorter, and the physical property it uses is sound.

2 What a cell is, to a physicist

Strip away the biology for a moment. To the equations in this software, a cell is:

A sphere	of radius r , between $1\ \mu\text{m}$ (a platelet) and $10\ \mu\text{m}$ (a tumour cell). A human hair is about $70\ \mu\text{m}$ across.
Slightly heavier than water	density ρ_p between $1050\ \text{kg/m}^3$ and $1100\ \text{kg/m}^3$, where water is $997\ \text{kg/m}^3$. So 5–10 % heavier.
Slightly harder to squeeze than water	compressibility κ_p around $3.7 \times 10^{-10}\ \text{1/Pa}$, where water is $4.48 \times 10^{-10}\ \text{1/Pa}$. So about 20 % stiffer.

That is the entire physical description. Three numbers: how big, how heavy, how stiff. Everything this simulation does with acoustics follows from those three and nothing else.

Where this stops being true

A real cell is not a sphere of uniform jelly. It has a membrane, a nucleus that is denser than the rest, and a shape that changes when you push it. Treating it as a uniform sphere is an approxi-

mation that works well for the acoustic question and badly for others. Section 24 lists what this costs.

The specific cells in the default simulation

Key	Cell	r [μm]	ρ_p [kg/m^3]	κ_p [10^{-10} 1/Pa]	Role
mcf7	MCF-7 breast cancer	9.00	1068	3.72	the target
rbc	red blood cell	2.78	1099	3.41	the background
wbc	white blood cell	4.25	1070	3.99	background
platelet	platelet	1.19	1058	3.13	background
<i>water, for comparison</i>		—	997	4.48	

MCF-7 is a line of breast cancer cells taken from one patient in 1970 and grown in laboratories ever since. It is used here because it is the line acoustic separation papers use, so its measured properties exist in the literature.

In plain English

Notice how similar the numbers are. A tumour cell and a red blood cell differ by 3 % in density and 9 % in stiffness. If those were the only differences, no sorter could work. What saves us is the first column: the tumour cell is 3.2 times wider, and as we will see, width is the property that gets raised to the third power.

Cells are not all the same size

Real populations have a spread. A “9 μm ” tumour cell means a distribution centred on 9 μm . The software draws each simulated cell’s radius from a log-normal distribution — the standard empirical shape for cell-size histograms, and one that cannot produce a negative radius.

$$r \sim \text{LogNormal}(\mu, \sigma), \quad \sigma^2 = \ln(1 + \text{CV}^2), \quad \mu = \ln \bar{r} - \frac{1}{2}\sigma^2. \quad (1)$$

$\bar{r}$ the average radius you want, e.g. 9 μm

CV the spread, as a fraction of the average: 0.12 means $\pm 12\%$

μ, σ the parameters the log-normal formula actually needs; the two lines above convert from what you know to what it wants

This matters more than it looks. Because the sorting force depends on r^3 , a cell 12 % smaller than average feels $0.88^3 = 68\%$ of the average force. The spread in cell size is the main reason a sorter is never 100 % pure.

Part II — The trick, which is physics

3 Sound is squeezing

Sound is not a substance travelling through water. It is a pattern of squeezing and stretching that travels through it. At any point in the liquid, the pressure goes slightly up, then slightly down, then up again, millions of times a second.

Three numbers describe it:

- f frequency — how many times a second the squeezing repeats. Ours is 6.632 MHz: 6.6 million times a second, about 300
- c speed of sound — how fast the pattern travels. 1497 m/s in water, four times faster than in air.
- λ wavelength — the distance between one squeeze and the next, $\lambda = c/f$.

In plain English

Frequency is how *fast* it wiggles. Wavelength is how *far apart* the wiggles are. They are locked together by the speed of sound: pick one and the other follows.

4 Standing waves: the part where nothing moves

Send a wave along a channel and let it bounce back. The outgoing and returning waves add up. In some places they always cancel; in others they always reinforce. The pattern stops travelling and just sits there, pulsing in place. That is a standing wave.

The picture to hold in your head

Hold one end of a skipping rope and have a friend hold the other, then shake it at the right rate. The rope settles into a shape where some points thrash up and down violently while other points — the *nodes* — never move at all. The rope is not going anywhere. It is standing. Sound in the channel does the same thing, except what is oscillating is pressure, not height. There are places where the pressure never changes (pressure nodes, the quiet places) and places where it swings hardest (pressure antinodes, the loud places).

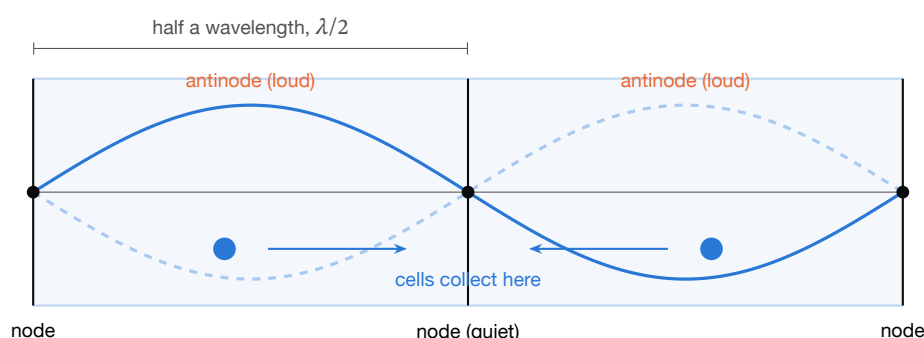


Figure 1: A standing pressure wave across the channel. The solid and dashed curves are the two extremes of the pressure swing. At the nodes the pressure never changes; at the antinodes it swings hardest. Cells that are denser and stiffer than water — which is every cell in the body — drift towards the nodes.

The spacing between neighbouring nodes is always *half* a wavelength. That single fact is what lets us design the device: choose the wavelength, and you have chosen where the cells will collect.

5 Why a standing wave pushes things

Here is the part that is genuinely surprising. The pressure at a node never changes. So why would a cell sitting slightly off a node feel any push at all?

The answer is that the cell *scatters* the sound. It is denser and stiffer than the water around it, so it does not move quite in step with the liquid. It re-radiates a small wave of its own, and that scattered wave interferes with the original. When you average over one full cycle of the oscillation — which takes 150 nanoseconds, so “averaged” is a fair description of anything the cell experiences — the interference does not cancel. It leaves behind a small, steady push.

The picture to hold in your head

Imagine an invisible landscape of hills and valleys laid over the channel. A cell placed on it rolls downhill, exactly like a marble. It is not being blown along by the sound; it is sitting on a slope that the sound created.

The technical name for this landscape is the Gor'kov potential, after the Soviet physicist who worked it out in 1962. Where the valleys are — at the quiet nodes or at the loud antinodes — depends on the cell.

The height of this landscape at any point is:

$$U = V_c \left[\underbrace{\frac{f_1}{2} \kappa_f \langle p^2 \rangle}_{\text{from squeezing}} - \underbrace{\frac{3}{4} f_2 \rho_f \langle v^2 \rangle}_{\text{from sloshing}} \right] \quad (2)$$

U	the height of the landscape at that point [joules]
V_c	the cell's volume, $\frac{4}{3}\pi r^3$ [m ³]
$\langle p^2 \rangle$	how hard the pressure swings there, averaged over a cycle
$\langle v^2 \rangle$	how hard the water sloshes back and forth there, averaged
ρ_f, κ_f	the water's density and compressibility
f_1, f_2	two numbers describing how much the cell differs from water — next section

And the force is simply how steep the landscape is:

$$\boxed{F = -\nabla U} \quad (3)$$

The ∇ (“nabla” or “grad”) just means *slope*. The minus sign means downhill. That is the whole of it: build the landscape, measure its slope, and you have the force.

In plain English

The sound field creates an invisible hilly surface. Cells roll downhill on it. The two terms in equation (2) are the two reasons a cell is not the same as the water it displaced: it resists being squeezed differently (first term), and it resists being shaken differently (second term).

6 The contrast factor: two questions, one number

The cell only has to answer two questions.

Question	The number	For a typical cell
Am I harder to squeeze than water?	$f_1 = 1 - \frac{\kappa_p}{\kappa_f}$ (the <i>monopole</i> term)	$f_1 = 1 - \frac{3.72}{4.48} = 0.169$, so yes, a bit
Am I heavier than water?	$f_2 = \frac{2(\rho_p - \rho_f)}{2\rho_p + \rho_f}$ (the <i>dipole</i> term)	$f_2 = \frac{2(1068-997)}{2136+997} = 0.045$, so yes, slightly

Combine them into a single acoustic contrast factor:

$$\Phi = f_1 + \frac{3}{2}f_2 = \frac{5\rho_p - 2\rho_f}{2\rho_p + \rho_f} - \frac{\kappa_p}{\kappa_f} \quad (4)$$

- $\Phi > 0$ the cell is denser *and* stiffer than water. It goes to the node. Every cell in the human body does this.
 $\Phi < 0$ the cell is lighter or squishier. It goes to the antinode. Fat droplets and gas bubbles do this.
 $\Phi = 0$ the object is acoustically indistinguishable from water, and nothing happens.

Particle	Φ	Where it goes
platelet	+0.360	node
red blood cell	+0.334	node
MCF-7 tumour cell	+0.237	node
white blood cell	+0.179	node
fat droplet	−0.273	antinode

The fat droplet is in the table on purpose: it is the test case. Any correct implementation of equation (4) *must* return a negative number for it, and the automated test suite checks exactly that.

Where this stops being true

The three-times trap. Two definitions of Φ circulate in the published literature and they differ by a factor of exactly three. The one above is the classical version. Some authors (including the most widely-cited modern tutorial) use $\Phi_{\text{alt}} = f_1/3 + f_2/2 = \Phi/3$, together with a force formula that has a compensating factor.

Mix them up and every force you compute is wrong by $3\times$ — and it will look plausible, because it is still the right order of magnitude. The software provides both under different names and the test suite pins their ratio at exactly 3.

7 The one equation the whole device rests on

For a one-dimensional standing wave, doing the slope calculation on equation (2) gives:

$$F_r(x) = -\frac{\pi p_0^2 V_c \kappa_f}{2\lambda} \Phi \sin(2k(x - x_{\text{node}})) \quad (5)$$

F_r	the sideways push on the cell [newtons]
x	where the cell is, measured across the channel
x_{node}	where the nearest quiet spot is
p_0	how loud the sound is (pressure amplitude) [pascals]
V_c	the cell's volume = $\frac{4}{3}\pi r^3$
κ_f	how squeezable water is
λ	the wavelength of the standing pattern
k	= $2\pi/\lambda$, just wavelength written the way physicists prefer
Φ	the contrast factor from equation (4)

Read it one piece at a time.

- $\sin(2k(x - x_{\text{node}}))$ — the direction. It is zero exactly at the node and exactly at the antinode, positive on one side and negative on the other. So the cell is always pushed towards the node, and once it arrives, the force switches off. It parks there.

- p_0^2 — turn the sound up twice as loud and the push quadruples. Not doubles. This is why the drive voltage is such an effective knob.
- $V_c \propto r^3$ — *this is the whole device*, and it gets its own section.
- Φ — which way and how strongly the cell responds.
- $1/\lambda$ — a tighter pattern pushes harder, because the landscape has steeper slopes.

8 Why size wins: the r^2 law

The cell is not in a vacuum. It is in water, and water resists. The resistance is Stokes drag:

$$F_d = 6\pi\mu r(u_f - u_p) \quad (6)$$

μ how thick the liquid is (viscosity); water is 0.89×10^{-3} Pa s
 r the cell's radius — note: radius, not volume
 $u_f - u_p$ how fast the cell is moving relative to the water around it

Now put the two together. The cell speeds up until the push and the drag balance, which happens almost instantly (see §11). At balance:

$$\underbrace{\text{push} \propto r^3}_{\text{equation (5)}} = \underbrace{\text{drag} \propto r \times \text{speed}}_{\text{equation (6)}} \implies \boxed{\text{speed} \propto r^2} \quad (7)$$

In plain English

The push grows with the cell's *volume*. The resistance grows only with its *radius*. Volume beats radius. Double the radius and the cell moves four times faster.

The picture to hold in your head

Drop a marble and a cannonball into a swimming pool. Both feel gravity pulling on their whole volume and water resisting only across their width. The cannonball wins, and not by a little. Acoustic sorting is exactly this, with sound instead of gravity, sideways instead of down.

Put the real numbers in. Comparing an MCF-7 tumour cell to a red blood cell:

$$\begin{aligned} \frac{\text{speed of MCF-7}}{\text{speed of RBC}} &= \left(\frac{r_{\text{MCF-7}}}{r_{\text{RBC}}} \right)^2 \times \frac{\Phi_{\text{MCF-7}}}{\Phi_{\text{RBC}}} \\ &= \left(\frac{9.00}{2.78} \right)^2 \times \frac{0.179}{0.252} = 10.5 \times 0.71 = 7.4 \end{aligned} \quad (8)$$

The tumour cell crosses the channel 7.4 times faster than the red cell. Give them both the same amount of time — which is what a fixed channel length does — and the tumour cell arrives at the middle while the red cell is still most of the way out.

Note that the contrast factor actually works *against* us — the red cell has a higher Φ (0.252 vs 0.179), because it is denser and stiffer relative to water than a tumour cell is. Size overwhelms it anyway, ten to one.

9 Where the sound comes from: an earthquake on a chip

We need a standing sound wave inside a channel $300 \mu\text{m}$ wide. There is no room for a loudspeaker. Instead, the channel sits on top of a slab of lithium niobate, a crystal that is *piezoelectric*: apply a voltage and it physically deforms; deform it and it produces a voltage.

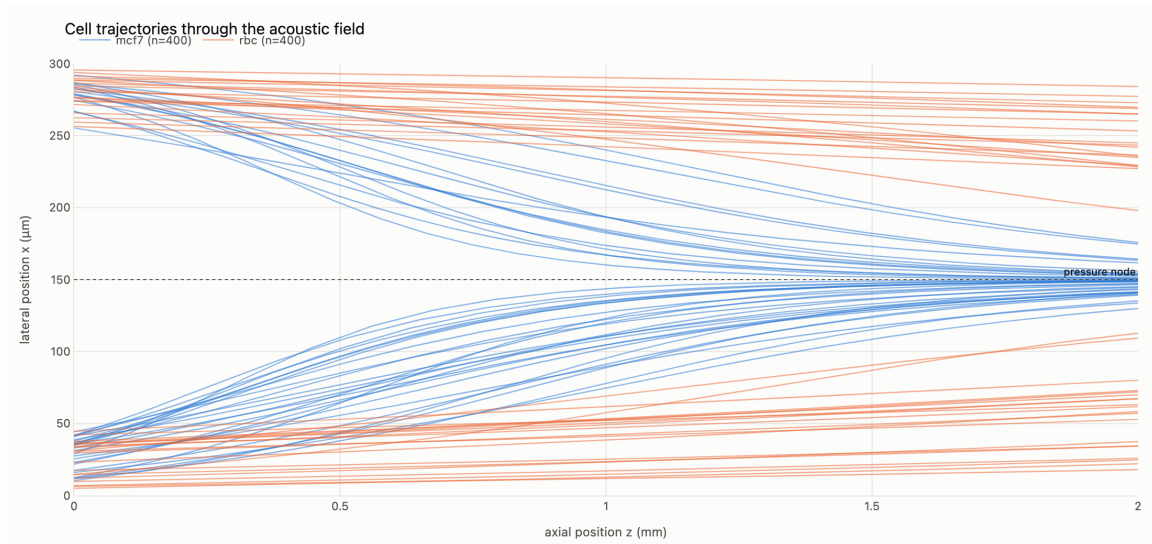


Figure 2: The r^2 law, simulated. 400 tumour cells (blue) and 400 red blood cells (orange) enter at the two walls and travel 2 mm down the channel, which takes 0.36 s. The tumour cells reach the middle; the red cells barely move. Nothing else is going on in this device.

On the surface of that crystal we lay down two interlocking metal combs, called an interdigital transducer or IDT. Apply an alternating voltage across the combs, and the crystal ripples — a tiny, extremely fast earthquake confined to the top surface. This is a surface acoustic wave.

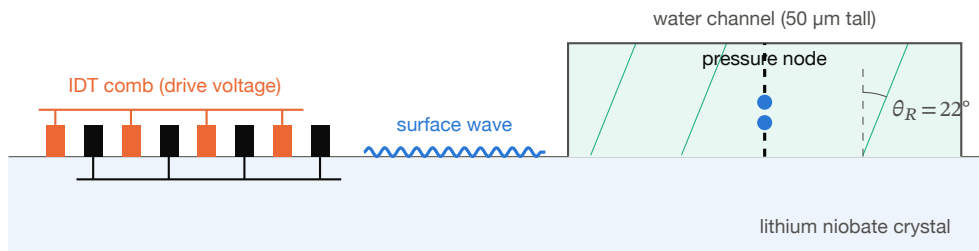


Figure 3: The device in cross-section. A comb-shaped electrode drives a ripple along the crystal surface. Because the ripple travels faster in the crystal (3979 m/s) than sound does in water (1497 m/s), it leaks into the liquid at a fixed angle of about 22° — the same reason a straw looks bent in a glass of water. Two such waves sent from opposite sides make the standing pattern.

The leak angle is set by the same rule that bends light entering water:

$$\theta_R = \arcsin\left(\frac{c_{\text{water}}}{c_{\text{SAW}}}\right) = \arcsin\left(\frac{1497}{3979}\right) = 22.1^\circ \quad (9)$$

We do not put this angle into the simulation. We tell the computer what the crystal surface is doing, ask it to solve the equation for sound in water, and the 22° comes out on its own. That is a small but real check that the physics is being solved rather than assumed.

10 Two things the textbook formula gets wrong here

Building this software turned up two places where applying the standard equations naively gives the wrong answer. Both are now handled explicitly, and both are the sort of thing that would quietly ruin a device design.

10.1 The ruler is the crystal, not the water

The obvious thing to assume is that the sound in the channel has the wavelength of sound in water: $\lambda_{\text{water}} = c_{\text{water}}/f$. At 20 MHz that is $75 \mu\text{m}$, so nodes every $37 \mu\text{m}$.

That is wrong. The pattern in the liquid is stamped onto it by the crystal underneath, so it inherits the crystal's spacing, not the water's:

$$\lambda_{\text{SAW}} = \frac{c_{\text{SAW}}}{f}, \quad \text{node spacing} = \frac{\lambda_{\text{SAW}}}{2} \quad (10)$$

At 20 MHz, $\lambda_{\text{SAW}} = 3979/20 \times 10^6 = 199 \mu\text{m}$, so nodes every $99.5 \mu\text{m}$ — nearly three times further apart than the naive answer.

Frequency	λ_{SAW}	Node spacing	Nodes in a $300 \mu\text{m}$ channel
6.632 MHz	$600 \mu\text{m}$	$300 \mu\text{m}$	1
13.26 MHz	$300 \mu\text{m}$	$150 \mu\text{m}$	2
20 MHz	$199 \mu\text{m}$	$99.5 \mu\text{m}$	3

Where this stops being true

This is not a detail. A sorter with one node in the channel collects everything into the middle and you tap the middle outlet. A sorter with three nodes collects cells into three separate stripes, each cell going to whichever stripe it started nearest. The outlet histogram becomes three-humped and the two-outlet design simply does not work.

The original specification for this project asked for 20 MHz in a $300 \mu\text{m}$ channel. That combination has three nodes. The software now detects this and prints the frequency that gives one:

$$f = \frac{c_{\text{SAW}}}{2W} = \frac{3979}{2 \times 300 \times 10^{-6}} = 6.632 \text{ MHz}$$

which is the default in the shipped configuration.

10.2 The sound does not travel straight across

The second correction is subtler. The standard force formula, equation (5), quietly assumes the wave travels purely *across* the channel. In this device it does not: the crystal sets the sideways spacing, and the laws of sound in water then force the wave to have an upward component as well. That upward component is precisely the 22° leak.

Writing k_x for the sideways wavenumber (set by the crystal) and k_f for what sound in water would have, they are related by Pythagoras:

$$k_x^2 + k_f^2 = k_f^2, \quad k_x = \frac{2\pi}{\lambda_{\text{SAW}}}, \quad k_f = \frac{2\pi f}{c_{\text{water}}} \quad (11)$$

Redoing the Gor'kov calculation with the tilt included gives a modified contrast factor:

$$\Phi_{\text{eff}} = f_1 + \frac{3}{2} f_2 \left(\frac{k_x}{k_f} \right)^2 \quad (12)$$

The squeezing term is untouched; the sloshing term is scaled down, because the water is sloshing partly up-and-down rather than side-to-side, and only the side-to-side part contributes to sideways force. When $k_x = k_f$ — a wave that really does travel straight across — the correction disappears and the textbook formula returns exactly.

Cell	Φ (textbook)	Φ_{eff} (this device)	Change
MCF-7	0.2371	0.1787	−25 %
red blood cell	0.3341	0.2519	−25 %

A 25 % error in force is a 25 % error in how far every cell travels, which near the edge of a collection outlet is the difference between a working device and a broken one. The correction is verified in the test suite by computing the force numerically from an exact solution of the wave equation and comparing: they agree to seven parts in a million.

11 Life at low Reynolds number: water is honey

At the scale of a cell, water does not behave the way your intuition says. The governing quantity is the Reynolds number, a pure ratio with no units:

$$\text{Re} = \frac{\rho U L}{\mu} = \frac{\text{how much the fluid wants to keep going}}{\text{how much it wants to stop}} \quad (13)$$

Situation	Re	What it feels like
A person swimming	10^6	you glide when you stop stroking
A goldfish	10^4	still glides
Our channel	0.53	no gliding at all
A cell in our channel	0.15	—
A bacterium swimming	10^{-5}	like swimming in tar

The picture to hold in your head

Stop pushing a cell and it does not coast. It stops within about 50 nm — less than one hundredth of its own diameter — in about 20 μs . There is no momentum, no drifting, no overshoot. The cell's velocity is determined entirely by the forces acting on it *right now*.

This is enormously convenient. Newton's second law, $F = ma$, is a differential equation in acceleration and needs two starting conditions. But when there is no meaningful acceleration, it collapses to something you can almost do in your head:

$$\underbrace{u}_{\text{cell speed}} = \underbrace{u_f}_{\text{speed of the water}} + \underbrace{\frac{F}{6\pi\mu r}}_{\text{extra speed from the push}} \quad (14)$$

The simulation uses this by default. It also implements the full $F = ma$ version, and the test suite checks that the two agree — which is how we know the approximation is safe here rather than merely convenient.

Where this stops being true

This breaks if the flow is fast enough or the particles large enough that Re approaches 1. The software computes both Reynolds numbers on every single run and prints a warning if either exceeds 1, rather than silently producing a number that looks fine and is wrong.

12 How the liquid flows: the parabola

The cells are carried along the channel by liquid pushed from a syringe pump. That flow is not uniform: it sticks to the walls (the “no-slip” condition, which is why blowing on a dusty surface never quite cleans it) and moves fastest in the middle.

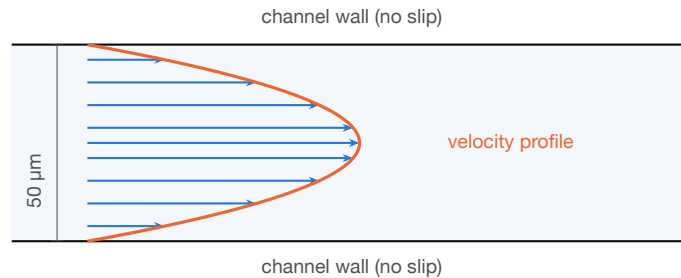


Figure 4: Pressure-driven flow in a channel. Zero at the walls, fastest in the middle. A cell near a wall takes much longer to travel the channel than one in the middle — which is why the simulation has to integrate long enough for the slowest cell, not the average one.

For a rectangular channel there is an exact answer, written as an infinite sum:

$$u_z(x, y) = \frac{16b^2G}{\mu\pi^3} \sum_{n=1,3,5,\dots} \frac{(-1)^{(n-1)/2}}{n^3} \left[1 - \frac{\cosh\left(\frac{n\pi x}{2b}\right)}{\cosh\left(\frac{n\pi a}{2b}\right)} \right] \cos\left(\frac{n\pi y}{2b}\right) \quad (15)$$

- u_z how fast the liquid moves along the channel at the point (x, y)
- a, b half the channel’s width and height
- G the pressure drop per metre pushing the liquid along
- n the sum index; each term is a smaller correction than the last

In plain English

This looks alarming and is not. It is a recipe: add up a few wavy shapes with decreasing weights and you get the parabola-like profile in the figure. Thirty terms are enough for six-digit accuracy. The software uses the exact sum rather than solving the flow numerically, because the exact answer is both faster and more accurate.

For the standard operating point — 5 $\mu\text{L}/\text{min}$ through a $300 \times 50 \mu\text{m}$ channel — this gives:

average speed	5.56 mm/s
peak speed (centre)	9.31 mm/s
time to cross the 2 mm active region	0.36 s
pressure needed	26.5 kPa/m

That 0.36 s is the entire budget. Every cell you want to collect has to reach the middle of the channel in 0.36 seconds, and every cell you want to discard has to fail to. Everything in the design — the frequency, the voltage, the flow rate, the length — is in service of that one race.

Part III — The second instrument: counting cells with electricity

13 A cell is a tiny insulator

The sorter answers “which cells are these?”. A different question a laboratory asks constantly is “how many cells are there, and are they still alive?” — for example, every hour for two days while testing whether a drug kills them.

Photographing the dish every hour and counting by eye is slow and disturbs the cells. There is a much better trick, and it relies on a fact about cell membranes:

In plain English

The membrane around a cell is a double layer of fat molecules about 5 nm thick. Fat does not conduct electricity. So a cell is, to a first approximation, a bag of salt water — which conducts well — wrapped in an insulating skin.

Put gold electrodes in the bottom of the dish and pass a tiny alternating current. When the dish is empty, the current flows freely through the salty growth medium. As cells settle and spread on the electrodes, they get in the way: the current has to squeeze through the narrow gaps between the cells and through the even narrower gap underneath them.

The picture to hold in your head

It is like laying cling film over a wire mesh at the bottom of a stream. The water has to find its way around the edges. The more cling film, the more the flow is obstructed — and the obstruction is something you can measure from outside without touching the water.

More cells $\Rightarrow$ more obstruction $\Rightarrow$ higher impedance. Impedance is just electrical resistance for alternating current, with one extra feature: it has a size *and* a timing shift, so it is written as a complex number Z .

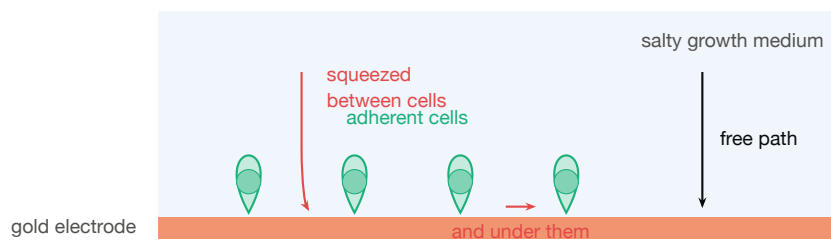


Figure 5: Electric cell-substrate impedance sensing. Current takes the easy route where the electrode is bare and has to squeeze through gaps where cells sit. The measured impedance therefore reports how much of the electrode is covered.

14 Turning impedance into a growth curve

The full relationship between how covered the electrode is and what the meter reads was worked out by Giaever and Keese in 1991. It is not pretty:

$$\frac{1}{Z_c} = \frac{1}{Z_n} \left[\frac{Z_n}{Z_n + Z_m} + \frac{Z_m/(Z_n + Z_m)}{\frac{\gamma I_0(\gamma)}{2I_1(\gamma)} + R_b \left(\frac{1}{Z_n} + \frac{1}{Z_m} \right)} \right], \quad \gamma = \alpha \sqrt{\frac{1}{Z_n} + \frac{1}{Z_m}} \quad (16)$$

Z_c	what you measure when cells are present
Z_n	what you measure when the electrode is bare
Z_m	the resistance of the cell membranes themselves
R_b	how tightly the cells are sealed to each other — “tight junctions”
α	how narrow the gap is between the cell’s underside and the gold
I_0, I_1	Bessel functions — standard mathematical functions that appear whenever something spreads out radially from a po

In plain English

Do not read the formula. Read what goes into it: how bare is the electrode, how insulating are the membranes, how tightly are the cells stuck to each other, and how thin is the gap underneath them. Those four things determine the reading. The equation is just the bookkeeping.

Laboratory instruments report a simplified, unitless version called the Cell Index:

$$CI(t) = \frac{|Z_{\text{cells}}(t)| - |Z_{\text{background}}|}{Z_{\text{reference}}} \quad (17)$$

Zero means an empty electrode. It climbs as the cells attach, spread and divide. It falls when they die and round up.

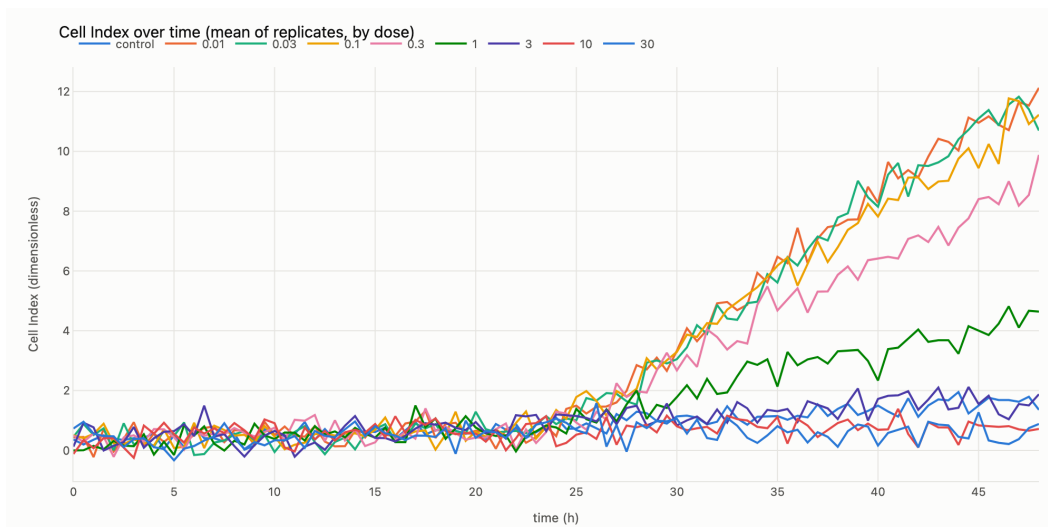


Figure 6: Simulated Cell Index over 48 hours, one line per drug dose. All wells grow identically for the first 24 hours; the drug is added at hour 24 and the curves separate by dose. This is the raw output of a real instrument, and this plot is what the simulation reproduces.

15 IC50: the number a drug is judged by

Take the Cell Index at the end of the experiment, plot it against drug concentration, and you get an S-shaped curve on a logarithmic axis: no effect at low dose, everything dead at high dose, and a transition in between. Fit it with the standard four-parameter logistic:

$$y = y_{\min} + \frac{y_{\max} - y_{\min}}{1 + (c/IC_{50})^h} \quad (18)$$

c	drug concentration
y	how much cell signal is left
IC_{50}	the answer: the concentration that kills half of them
h	the Hill slope — how sharp the switch from “alive” to “dead” is

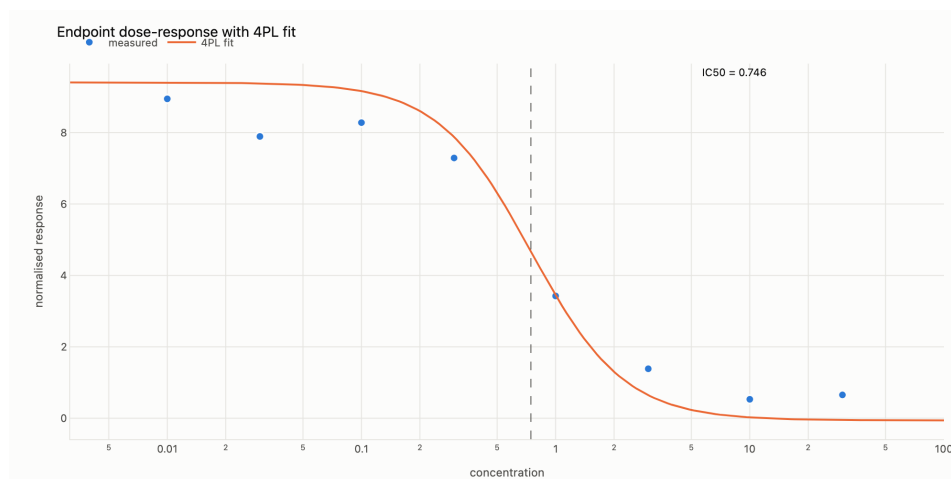


Figure 7: Dose–response from the simulated plate, with the fitted curve and the extracted IC50.

Where this stops being true

An honest wrinkle the simulation reproduces. If you plant a true IC50 of 1.0 in the simulation and then measure it from the endpoint, you get 0.75. That is not a bug.

The drug acts for 24 hours against cells that take about 20 hours to double, so the population has not finished responding when you stop the experiment. The measured IC50 always drifts towards its true value as exposure lengthens. Real endpoint assays do exactly this, which is why published IC50 values are meaningless without the exposure time attached.

Part IV — The third and fourth instruments: looking at pictures

16 Finding cells in an image

Given a photograph down a microscope, how does a computer count the cells?

The naive answer — “anything brighter than a threshold is a cell” — fails on real images for three reasons, and each step of the pipeline exists to undo one of them:

Problem	Why it happens	The fix
The image is brighter in the middle than the corners	the lamp is not perfectly even	subtract a heavily blurred copy of the image from itself
The image is speckly	light arrives as discrete photons, and counting anything random has $\sqrt{N}$ noise	blur slightly
Touching cells merge into one blob	they are genuinely touching	watershed, below

The picture to hold in your head

Watershed. Read the image as a landscape where bright means high, then turn it upside down so each cell becomes a basin. Now flood the whole terrain from the bottom of every basin at once. Where two floods meet, build a wall. Those walls are the boundaries between touching cells. The name is literal: it is the same computation a geographer does to work out which river a raindrop will end up in.

The software also generates its own test images, with the true answer known exactly, so the pipeline can be *scored* rather than admired. On a synthetic field of 150 cells it recovers 126. The 24 it misses are cells touching the frame edge (deliberately discarded — a cell cut in half by the edge has no measurable size) and a few genuinely merged pairs.

17 Counting, and why 100 cells means $\pm 10\%$

Once cells are found, concentration is geometry. A standard counting chamber has a lid held exactly 0.1 mm above the floor and a 1 mm grid ruled on it, so one grid square holds exactly 0.1 μL :

$$\text{concentration} \left[\frac{\text{cells}}{\text{mL}} \right] = \frac{N}{\text{area} \times \text{depth}} \times \text{dilution} \quad (19)$$

There is a statistical catch that matters more than any software detail. Cells land in the counting square at random, so the count follows Poisson statistics and carries an unavoidable uncertainty:

$$\frac{\text{uncertainty}}{\text{count}} = \frac{1}{\sqrt{N}} \quad (20)$$

Cells counted	Uncertainty	
25	$\pm 20\%$	
100	$\pm 10\%$	<i>a typical hemocytometer count</i>
400	$\pm 5\%$	
2500	$\pm 2\%$	

In plain English

If you count 100 cells and your colleague counts 100 cells from the same tube, you should expect to differ by about 10 % *even if you both do everything perfectly*. That is usually larger than the difference people are trying to detect. The software reports this number alongside every count, because leaving it out is how people end up believing differences that are not there.

Alive or dead

Trypan blue is a dye that a healthy membrane keeps out and a dead one lets in. Dead cells go dark. So viability is just the fraction of detected objects that are *not* dark, and the software splits the population by brightness.

18 Following cells over time

Take a picture every ten minutes for four hours and you can watch cells crawl. Two computational problems arise.

Problem one: which cell is which? Between two frames every cell has moved. The computer must decide that the cell at position A in frame 5 is the same cell as the one at position B in frame 6. The standard method assumes each cell moves less than some distance and picks the assignment that minimises total movement.

Where this stops being true

This one parameter — the maximum allowed step — is the whole game. Set it too small and fast cells appear to vanish and be replaced. Set it too large and the computer starts swapping the identities of cells that pass near each other. There is no automatic right answer; the software exposes it as `search_range_px` rather than hiding a guess.

Problem two: what does a track mean? A wiggly line is not a result. The standard summary is the mean-squared displacement: for every pair of frames separated by a time gap τ , how far did the cell get?

$$\text{MSD}(\tau) = \langle |r(t + \tau) - r(t)|^2 \rangle = 4D \tau^\alpha \quad (21)$$

The exponent α is the informative part:

α	Meaning	Everyday version
≈ 1	random diffusion	a drunk stumbling: distance ² grows with time
≈ 2	straight-line motion	someone walking somewhere: distance grows with time
< 1	confined or crowded	pacing in a small room

In plain English

A drunk who takes one step per second in a random direction ends up $\sqrt{N}$ steps from the pub after N seconds — so distance² grows *linearly* with time, giving $\alpha = 1$. Someone who actually walks somewhere covers N steps, so distance² grows as time *squared*, giving $\alpha = 2$. Measuring α tells you which kind of cell you are looking at, and cancer cells becoming more directional is exactly what invasion looks like.

The synthetic test video is built from cells with a known persistence, and the software recovers $\alpha = 1.89$ — correctly reporting mostly-directed motion — and a speed of $0.1446 \mu\text{m}/\text{min}$ against a planted 0.1430 : a 1 % error. That agreement is the evidence the tracking code works.

Part V — How a computer actually solves any of this

19 Why you cannot do this with a pen

The equation for sound in the channel is:

$$\nabla^2 p + k^2 p = 0 \quad (22)$$

p the pressure, at every single point in the channel
 $\nabla^2 p$ how much the pressure at a point differs from the average of its neighbours — a measure of curvature
 k $2\pi/\lambda$, set by the frequency

In plain English

Equation (22) says: *at every point, the curvature of the pressure must be proportional to the pressure itself*. It is one rule that has to hold simultaneously everywhere — and “everywhere” is infinitely many places. That is why it needs a computer.

For a plain rectangle you can solve it by hand. Add a rubber wall on one side, a vibrating crystal on the bottom, and an inlet that is not quite square, and you cannot.

20 Finite elements: chop it into triangles

The trick is 80 years old and startlingly simple.

The picture to hold in your head

You cannot describe a curved hillside with one equation. But you can cover it with thousands of small flat triangles, each tilted a bit differently. Zoom out and the triangles look like a smooth hill. Zoom in and each one is trivially simple.

Do the same to the pressure field. Chop the channel into triangles, assume the pressure varies simply inside each one, and require that neighbouring triangles agree along their shared edges.

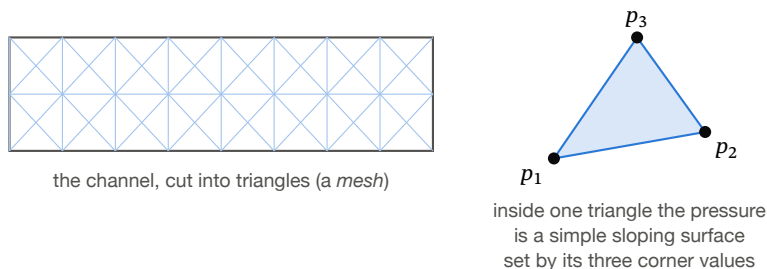


Figure 8: The finite element method in one picture. Unknown “the pressure everywhere” becomes unknown “the pressure at each corner”, which is a finite list of numbers a computer can solve for.

This converts an impossible calculus problem into a very large but ordinary algebra problem:

$$\mathbf{A} \mathbf{p} = \mathbf{b} \quad (23)$$

$\mathbf{p}$ the list of unknown pressures, one per corner — a few thousand numbers
 $\mathbf{A}$ a big table encoding “each corner must be consistent with its neighbours”
 $\mathbf{b}$ what is driving the system — here, the vibrating crystal underneath

Solving simultaneous equations is something computers are extremely good at. A typical run here has about 20 000 unknowns and takes under a second.

In plain English

Every finite element package in the world — the free ones in this project and the commercial ones costing tens of thousands per seat — does this same thing. They differ in convenience, geometry handling and support, not in the idea.

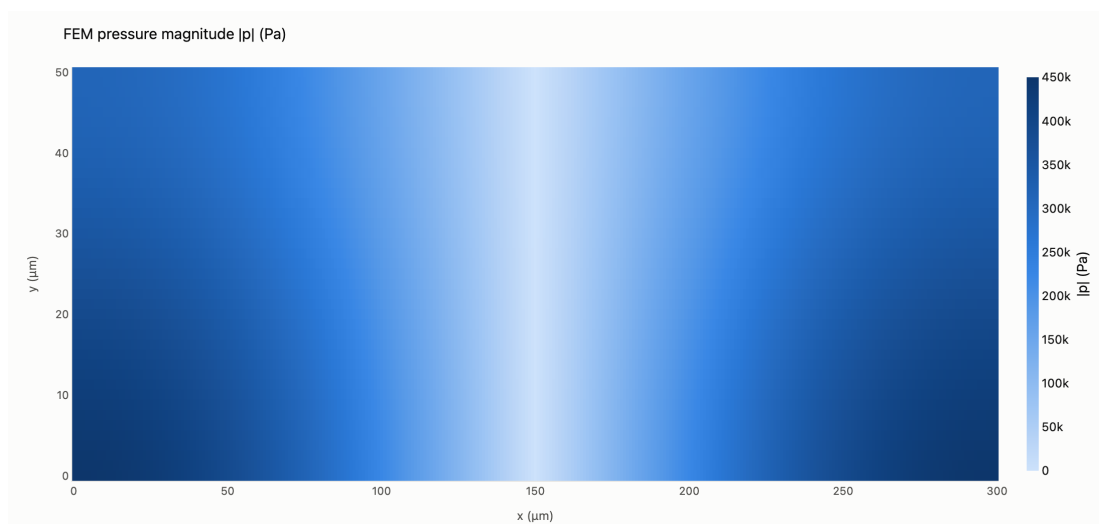


Figure 9: The computed pressure field in the channel cross-section. Dark blue is where the pressure swings hardest; the pale vertical stripe down the middle is the node, where it never changes at all — that is where the cells collect. The channel is 300 μm wide and 50 μm tall. Notice that the dark bands fade towards the top: the sound leaks out through the ceiling, and that fading turns out to matter a great deal (§24).

21 Moving the cells

With the force field known, each cell obeys equation (14): its velocity is the water’s velocity plus the acoustic push divided by the drag. Written as a rule for how position changes with time:

$$\frac{d\mathbf{x}}{dt} = \mathbf{u}_f(\mathbf{x}) + \frac{\mathbf{F}(\mathbf{x})}{6\pi\mu r} \quad (24)$$

This is an ordinary differential equation: a rule for the rate of change, from which the computer reconstructs the path.

The picture to hold in your head

It is dead reckoning. You know your speed and heading at every instant; step forward a tiny amount of time, recompute, repeat. The only skill is choosing the step size — small enough to be accurate, large enough to finish. The software lets the solver choose adaptively: small steps where the force changes quickly, large steps where it does not.

All the cells are integrated as one system rather than one at a time, which is why 800 cells take under a second.

One subtlety that cost real debugging time

Cells reaching a wall must not leave the channel. The obvious fix is to zero out any velocity pointing into the wall. Doing that made the simulation roughly a hundred times slower.

The reason is that it makes the rule for velocity *jump* discontinuously at the wall. Adaptive solvers detect a jump by taking smaller and smaller steps trying to resolve it, and effectively stall. The fix is to leave the rule smooth and instead evaluate the force at the nearest point *inside* the channel, then clip the final path. Same physical answer, no discontinuity, full speed.

In plain English

This is what most of the work in scientific computing actually is: not inventing physics, but noticing that a mathematically harmless choice is numerically catastrophic.

22 How do you know the code is right?

This is the question that matters, and there is only one honest answer: compare against cases where you already know the answer. A simulation that has never been checked against anything is a picture, not a result.

What is checked	Against what	Agreement
The electrical solver	the schoolbook formula for two parallel plates, $Z = L/(\sigma A)$	10^{-9}
The general force machinery	the hand-derived one-dimensional formula, on a field where both apply exactly	7×10^{-6}
The sound solver	the known shape of a standing wave in a box	$< 2\%$
The flow formula	numerically integrating its own answer to recover the flow rate it was given	$< 10^{-3}$
The flow formula, again	the exact textbook result for an infinitely wide channel (peak = $1.5 \times$ mean)	1.5047
The no-inertia shortcut	the full $F = ma$ calculation	$< 10^{-3}$
The IC50 fitter	synthetic data with a planted answer	$< 10^{-3}$
The tracking code	synthetic video with planted speed	1 %

Beyond numerical agreement, the test suite checks *identities* — things that must be true regardless of the numbers:

- The force is exactly zero at nodes and at antinodes.
- Doubling the loudness quadruples the force; doubling the volume doubles it.
- Every cell in the body has $\Phi > 0$; a fat droplet has $\Phi < 0$.
- The two rival definitions of Φ differ by exactly 3.
- No cell is ever lost, duplicated, or found outside the channel.
- Turning the sound off produces no separation at all.
- Raising the flow rate always reduces capture.

In plain English

Checking a number is weak evidence; checking a *relationship* is strong evidence. If someone changes the code and the force stops scaling with the cube of the radius, a test fails immediately — even though no specific number was ever written down.

There are 160 such checks. They run in 14 seconds and all pass, with no optional software installed.

Convergence: is the mesh fine enough?

A finite element answer depends on how finely you chop. If the answer keeps changing as you refine, you are looking at your mesh, not at physics. The symmetry error — which must be zero in reality — shrinks steadily:

Mesh resolution	Peak force	Symmetry error
32	109.60 pN	0.118
64	106.70 pN	0.021
128	106.23 pN	0.009

Both columns settle down. That is what convergence looks like, and it is the difference between a computed answer and a decorated guess.

Part VI — The numbers, and what is wrong with them

23 Every number in the default experiment

What we choose (the device)		
Frequency	f	6.632 MHz
Drive voltage	V_{pp}	15 V
Channel width	W	300 μm
Channel height	H	50 μm
Active length	L	2 mm
Flow rate	Q	5 $\mu\text{L}/\text{min}$
What follows from those (the physics)		
SAW wavelength	λ_{SAW}	600 μm
Node spacing	$\lambda_{\text{SAW}}/2$	300 μm
Number of nodes in the channel		1, at the centre
Leak angle	θ_R	22.1°
Sound pressure	p_0	0.45 MPa
Average flow speed	$\bar{u}$	5.56 mm/s
Time in the device	$L/\bar{u}$	0.36 s
Channel Reynolds number	Re	0.53
Particle Reynolds number	Re_p	0.15
What each cell feels		
Peak force on a tumour cell		129.6 pN
Peak force on a red cell		5.4 pN
Force ratio		24×
Speed ratio (force ÷ drag)		7.4×
Tumour cell's top speed sideways		858 $\mu\text{m}/\text{s}$
What comes out		
Recovery (tumour cells collected)		100 %
Purity (of what was collected)		94.3 %
Enrichment over the input ratio		1.9×

In plain English

129.6 pN is a piconewton — a trillionth of a newton. It is roughly the force gravity exerts on a single bacterium. It is also about ten times the force a molecular motor inside your cells produces, and about a hundredth of the force needed to break a chemical bond. It is small enough not to damage the cell and large enough to steer it. That window is why this technique exists at all.

On a realistic blood sample

The demonstration above uses a 50:50 mixture, which is convenient and not realistic. With tumour cells spiked at 1 % into a full blood background of red cells, white cells and platelets, the same device gives:

Recovery	100 %
Purity of the collected fraction	10 %
Enrichment	10×

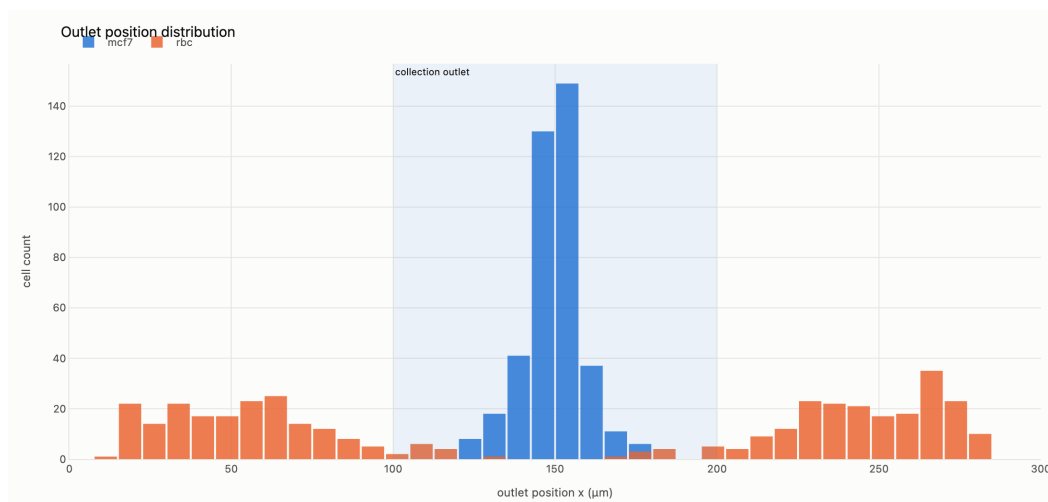


Figure 10: Where the cells come out. Blue tumour cells pile into the central collection outlet; orange red cells stay near the walls where they started. The small orange overlap in the middle is the 6 % of red cells that were unusually large, or started closer to the centre, and it is the reason purity is 94 % rather than 100 %.

Enrichment is the number that matters clinically: the sample handed to the next stage is ten times richer in tumour cells than the blood that went in, with none of them lost. Real workflows chain several such stages.

24 What is assumed, and what is wrong

This is the most important section in the document.

The single biggest assumption

Nothing in this software predicts how loud the sound is for a given drive voltage. That calculation requires solving the full electrical-mechanical behaviour of the crystal and the electrode pattern, which is a separate and harder problem.

Instead there is a straight-line calibration — 15 volts gives 0.45 MPa — which is inside the range published for devices of this kind, and is labelled in the source code as an assumption in capital letters. It scales every force in the simulation.

In plain English

If you build this device, do not trust the voltage. Measure the pressure — by tracking calibration beads of known properties and fitting how fast they move — and put that measured number in. The software accepts it directly for exactly this reason.

The simple model is optimistic

The one-dimensional formula, equation (5), applies the sound strength measured at the crystal surface to cells at *every* height in the channel. The real field weakens with height as it leaks out through the ceiling:

Height above the crystal	Sideways force
0 μm	106.7 pN (the value the simple model uses everywhere)
10 μm	88.5 pN
20 μm	64.9 pN
30 μm	41.9 pN
40 μm	25.6 pN
50 μm	19.9 pN (at the ceiling)
Average across the channel	53 % of the surface value

A cell drifting near the ceiling feels a fifth of the push the simple model assumes. So the simple model is a best case, and the finite element model is the realistic one. Both ship, the difference is measured by a script that anyone can run, and the test suite forbids the finite element result from ever coming out more optimistic than the simple one.

Things left out entirely

Left out	Why, and when it will bite you
Acoustic streaming	Sound loses a little energy at the walls, which stirs the liquid into slow steady swirls. This stirring does <i>not</i> depend on particle size, while the sorting force scales with r^3 . Below a critical radius — about 1 μm in water at these frequencies — the stirring wins and particles are mixed rather than sorted. Irrelevant for 9 μm tumour cells; fatal for bacteria or platelets.
Cells pushing each other	Two cells in the same sound field scatter onto each other and attract. This is why acoustophoresis videos show cells forming chains. Ignored because it costs N^2 computation and only matters above roughly 10^6 cells per mL.
Vertical forces	The model computes them but does not apply them by default. In a genuine channel, cells are held at a stable height by lift forces this two-dimensional model does not include; without those, cells pile against a wall where the flow is zero and never leave. Turning it on makes the failure visible rather than hidden — the results table reports that cells failed to exit.
Cells being squishy	A cell is treated as a rigid sphere of uniform jelly. Real cells deform, and deformability is itself a cancer marker. A different physics entirely.
The membrane and nucleus	For sound, the cell is one uniform blob. For the impedance instrument, the membrane is modelled explicitly, because there it is the whole effect.

Where the model is being stretched even when it works

The whole acoustic theory assumes the cell is much smaller than the sound wavelength. The measure of this is ka , roughly “how many wavelengths across is the cell”. The usual limit is $ka < 0.1$.

Frequency	ka for a 9 μm cell
6.632 MHz	0.39 beyond the textbook limit — warned
20 MHz	1.06 badly beyond it — warned loudly

Where this stops being true

The default operating point is already outside the strict validity of the theory it uses. This is not a flaw introduced here — the published experimental literature on acoustic tumour-cell separation operates in the same regime, and the theory is known to over-predict the force there.

The software’s response is to compute the number and print a warning naming the reference, on

every run, rather than to stay quiet. A model that tells you when it is being used outside its range is more useful than one that is silently narrower.

Where the numbers come from

Every physical constant in the software carries either a published reference (a DOI) or an explicit label saying it is an assumption together with the reason. There is no third category, and an automated test refuses to let anyone add one.

Of the 65 material values, 27 have published sources and 38 are flagged assumptions. Running `biosim materials` prints the 38, with the reason for each. That list is exactly what a methods section ought to disclose.

In plain English

The point is not that assumptions are bad. It is that a reader should be able to find out, in one command, which numbers were measured by someone and which were chosen by the author — rather than having all 65 presented with the same authority.

25 The whole thing on one page

1. The problem is a needle in a haystack. One tumour cell among five billion blood cells, and it must survive the search.
2. The handle is size. Tumour cells are about three times wider than red blood cells, and nothing a cell can do hides that.
3. Sound provides a gentle, invisible landscape. Two surface waves sent from opposite sides of a chip make a standing pattern with quiet lines and loud lines. Cells roll downhill towards the quiet lines.
4. The push grows with volume; the resistance grows with width. So sideways speed scales with radius squared. A cell three times wider moves about ten times faster. *This single fact is the device.*
5. Everything else is timing. Cells spend 0.36 seconds in the sound field. Tumour cells reach the middle in that time; red cells do not. Tap the middle outlet.
6. The maths is one equation with five factors: how loud, how big the cell, how different from water, how tight the pattern, and where the cell sits relative to the quiet line.
7. The computation is triangles and dead reckoning. Chop the channel into triangles to find the sound field; then step each cell forward in tiny time increments.
8. The credibility is in the checking. 160 automated tests compare against hand-solvable cases, verify relationships that must hold regardless of numbers, and confirm the answer stops changing as the mesh is refined.
9. The honesty is in the warnings. The model announces when it is outside its validity, when the operating point produces the wrong number of quiet lines, and which of its inputs were assumed rather than measured.
10. The same skeleton runs three more instruments — impedance monitoring, cell counting, cell tracking — because they share the same core: describe the experiment, validate it, solve it, save it, plot it.

A closing note on what simulation is for. None of this replaces an experiment. What it does is let you find out, in seconds and at no cost, that 20 MHz in a 300 μm channel gives you three collection stripes instead of one — rather than finding out after fabricating the chip. That is the whole value proposition: mistakes made in software are cheap.

A Every symbol, in one table

Symbol	Name	Plain meaning
r	radius	how big the cell is
V_c	volume	$\frac{4}{3}\pi r^3$
ρ_p, ρ_f	density	how heavy the cell / the water is, per unit volume
κ_p, κ_f	compressibility	how easily the cell / the water squeezes
μ	viscosity	how thick the liquid is
c	speed of sound	how fast the pattern travels
f	frequency	how many wiggles per second
λ	wavelength	distance between wiggles
k	wavenumber	$2\pi/\lambda$; wavelength, upside down
p_0	pressure amplitude	how loud
Φ	contrast factor	which way the cell goes, and how strongly
f_1, f_2	scattering terms	the stiffness part and the weight part of Φ
U	Gor'kov potential	the height of the invisible landscape
F_r	radiation force	the sideways push from the sound
F_d	drag force	the water's resistance
Re	Reynolds number	whether the liquid coasts (it does not, here)
Q	flow rate	how fast liquid is pumped in
Z	impedance	electrical resistance for alternating current
CI	Cell Index	how covered the electrode is
IC ₅₀	half-maximal dose	the drug concentration that kills half
MSD	mean-squared displacement	how far a cell wanders
α	MSD exponent	1 = random wandering, 2 = walking somewhere

B Units, for people who do not think in them

Unit	Size	Something that size
metre	1	a long stride
millimetre (mm)	one thousandth	thickness of a coin
micrometre (μm)	one millionth	a red blood cell is 8 across
nanometre (nm)	one billionth	a cell membrane is 5 thick
newton (N)	1	the weight of an apple
piconewton (pN)	one trillionth	the forces in this document
pascal (Pa)	1	very small pressure
kilopascal (kPa)	one thousand	atmospheric pressure is 101
megapascal (MPa)	one million	our sound pressure is 0.45
hertz (Hz)	1 per second	a slow heartbeat
megahertz (MHz)	one million per second	our frequency is 6.6
litre (L)	1	a carton of milk
microlitre (μL)	one millionth	a small droplet; we pump 5 per minute

C Where each idea came from

The physics here is not new. This software implements published results; the contribution is putting them together in open code with the assumptions labelled.

Idea	Source
The force on a small particle in a sound field	Gor'kov (1962); modern derivation in Bruus, <i>Lab on a Chip</i> 12:1014 (2012), doi:10.1039/c2lc21068a
Node spacing is set by the crystal, not the liquid	Shi et al., <i>Lab on a Chip</i> 9:3354 (2009), doi:10.1039/b910595f
Sorting tumour cells this way	Li et al., <i>PNAS</i> 112:4970 (2015), doi:10.1073/pnas.1504484112
Drag on a sphere in slow flow	Stokes (1851); Batchelor, doi:10.1017/CB09780511800955
Flow in a rectangular channel	White, <i>Viscous Fluid Flow</i> ; Bruus, <i>Theoretical Microfluidics</i>
Measuring cells with electrodes	Giaever & Keese, <i>PNAS</i> 88:7896 (1991), doi:10.1073/pnas.88.17.7896
Following particles between video frames	Crocker & Grier, <i>J. Colloid Interface Sci.</i> 179:298 (1996), doi:10.1006/jcis.1996.0217
Cell motion as a persistent random walk	Selmeczi et al., <i>Biophysical Journal</i> 89:912 (2005), doi:10.1529/biophysj.105.061150
Acoustic streaming, and when it takes over	Muller et al., <i>Lab on a Chip</i> 12:4617 (2012), doi:10.1039/c2lc40612h

The complete list of sources, including every material constant and every value that was assumed rather than measured, is in docs/physics.md and can be printed with `biosim materials`.